

PRODUCT REQUIREMENTS DOCUMENT

PlateLensAI

AI-Powered Visual Nutrition Feedback System for Nigerian Students

Product Name	PlateLensAI
Version	1.0
Date	April 2026
Status	In Testing
Platform	Mobile - iOS & Android (React Native)
Product POC (Point of Contact)	Inioluwa Adeye
Design POC	Inioluwa Adeye
Engineering POC	Omosebi Enoch
Marketing POC	
Approvers / Sign-off	Product Lead, Engineering Lead, Design Lead

1 . WHY - PROBLEM & OBJECTIVE

1.1 Market Context

Nigeria has over 2 million university students. Malnutrition, poor dietary habits, and nutrition-related illness are persistent and documented challenges in Nigerian tertiary institutions. Students live away from home, often for the first time, with limited cooking skills, low food budgets, and little to no nutritional awareness or structured guidance. Existing nutrition apps are built for Western diets and are completely irrelevant to the Nigerian food context; they do not recognise Jollof rice, Egusi soup, Suya, or Amala.

There is no mobile product in Nigeria that uses AI to give a student instant, visual, and personalised nutritional feedback on the actual meals they eat. PlateLensAI fills this gap.

1.2 Objectives

For the Business

- Build and establish PlateLensAI as the leading AI-powered nutrition product for Nigerian students.
- Demonstrate a feasible, scalable product model that can expand to secondary schools, hospitals, and corporate wellness.
- Generate user retention data to support future monetisation such as premium features and institutional licensing.
- Build a unique, exclusive dataset of Nigerian food images and nutritional data to establish a long-term competitive advantage.

For the Users

- Give Nigerian university students instant, accurate nutritional feedback on the meals they actually eat.
- Help students understand which food groups they are missing in their daily diet without requiring nutritional expertise.
- Build healthy eating awareness through consistent, non-judgmental, educational reminders.
- Make nutrition guidance feel personal, relevant, and Nigerian.

2 . HOW WE MEASURE SUCCESS

2.1 Associated Objectives and Key Results (OKRs)

OKR	Objective	Key Result
OKR 1	Establish PlateLensAI as the go-to nutrition app for Nigerian students	3,000 active users within 3 months of launch
OKR 2	Drive consistent daily engagement	40% of users analyze a meal at least 3 times per week
OKR 3	Deliver accurate and trusted AI feedback	AI food classification accuracy $\geq 85\%$ on test dataset
OKR 4	Build a sustainable retention loop	30-day retention rate $\geq 35\%$

2.2 Success Metrics

P - Primary Metric (North Star)

Weekly Active Meal Analysis, which is the total number of meal pictures analyzed per week across all active users. This single number tells us whether users are returning and whether the core AI feature is delivering value.

S - Secondary Metrics

- Daily Active Users (DAU) and Monthly Active Users (MAU).
- Average number of analysis per user per week.
- Nutrition Report completion rate, which is the percentage of users who view the full feedback after analyzing.
- Push notification open rate, which is an indicator for re-engagement effectiveness, showing how well users return when prompted.

G - Guardrail Metrics (Must not get worse)

- AI classification accuracy must not fall below 80%. Below this, user trust collapses.
- App crash rate must stay below 1% of sessions.
- Average analysis-to-result time must stay under 5 seconds.
- User-reported incorrect results must stay below 10% of analysis.

3 . USERS & PERSONAS

3.1 Primary Persona

Title	Details
Name	Temi, The Busy Undergrad
Age	21
Institution	University of Ibadan
Living situation	Student hostel
Daily food reality	Eats from campus canteens, local bukas, or cooks simple one-pot meals
Tech behaviour	Uses data efficiently and engages daily on Instagram, TikTok, and WhatsApp
Goal	Wants to stay healthy, feel good, and not fall sick but without spending money she doesn't have
Frustration	Has no idea if she is eating 'well', nobody ever taught her what balanced nutrition actually means in her context
Motivation to use PlateLensAI	Only needs to upload pictures, not manually log her meals

3.2 Secondary Persona

Title	Details
Name	Emeka, The Health-Conscious Postgrad
Age	27
Situation	Postgraduate student who is more intentional about healthy living
Goal	Wants to build a better eating routine, and understand what Nigerian foods actually provide
Frustration	International apps don't recognise his meals. MyFitnessPal has no entry for ogbono soup
Motivation	Wants data, trends and improvement over time.

3.3 Problems We Are Solving

Problem	User Need	How We Know It Exist
No nutrition feedback for Nigerian meals	Students need to know if their meals are balanced	Field interviews with 12 University of Ibadan students during project research phase
Western apps are irrelevant	Users need a tool that recognises local food	90% of popular nutrition apps have no Nigerian food database (validated by user testing)
No visible daily diet gaps	Users need to see which food groups they are missing	Students reported eating the same 2–3 foods repeatedly without awareness
Nutrition feels complex and academic	Users need simple, non-intimidating feedback	Existing resources (textbooks, NAFDAC guidelines) are inaccessible to average students

3.4 Jobs to Be Done

- When I eat a meal, **I want to know if it is nutritionally balanced, so I can make better choices tomorrow.**
- When I am about to cook or buy food, **I want to know what food groups I am missing today, so I can fill the gaps.**
- When I look back at my week, **I want to see a summary of my eating patterns, so I understand my habits.**

4. SOLUTION

4.1 Solution Brief

PlateLensAI is a React Native mobile application that allows Nigerian university students to take a picture of their meals or upload them and receive instant, AI-generated nutritional feedback. The app uses a fine-tuned MobileNetV2 transfer learning model trained on Nigerian food images to classify meals into the six major food groups - Carbohydrates, Proteins, Fats and oils, Vitamins, Minerals and Water; and then generates a personalised nutrition report and feedback for each meal. for each meal.

The product is designed to be effortless: one picture, one tap, instant result. No manual food logging, No calorie counting and No nutritional expertise required.

4.2 Alternatives Considered

Alternative	Why It is Considered	Why It is Rejected
Manual food logging (text input)	Lower technical complexity	Too much friction. Users abandon manual logging within days. Proven by MyFitnessPal retention data
Barcode scanner only	Faster to build	Packaged foods are a minority of Nigerian student meals. Most meals are homemade or canteen-cooked
Partner with a dietitian for manual reviews	Higher accuracy	Not scalable. Introduces delays. Breaks the instant feedback loop that drives retention
Build on top of an existing nutrition API	Faster time to market	No existing API covers Nigerian foods adequately, especially student meals. Would produce inaccurate results and destroy trust

4.3 Prioritization of Features

RICE (Reach, Impact, Confidence and Effort) Prioritization Table - Quantitative

Feature	Reach	Impact	Confidence	Effort	RICE Score	Priority
Meal picture analysis + AI Classification	5000	3 - massive	80%	3 - high	4000	P0 - Must Have
Nutrition Feedback Report	5000	3 - massive	80%	2 - medium	6000	P0 - Must Have
Meal History Log	4000	2 - moderate	70%	1 - small	5600	P1 - Should Have
Preferences and Push Notification Reminders	4000	2 - moderate	80%	2 - medium	4000	P1 - Should Have
Food Groups Dashboard (weekly view)	4000	2 - moderate	70%	3 - high	1866	P1 - Should Have
Social Sharing of Nutrition Score	2000	1 - minimal	50%	1 - small	1000	P2 - Nice to Have

MoSCoW Prioritization Table - Qualitative

Category	Features
Must Haves (essential features that need to be included)	Meal picture analysis + AI Classification, Nutrition Feedback Report, Settings Feature
Should Haves (important features we should try to include)	Meal History Log, Preferences and Push Notification Reminders
Would Haves (nice to have smaller improvements and additions)	Food Group Dashboard (weekly view), Social Sharing of Nutrition Score
Wants (features that do not match current needs or goals)	None identified at this stage

5 . PRODUCT FLOW & FEATURE DETAILS

5.1 Consumer Journey

Stage	What the User Does	What the App Does
1. Onboarding	Downloads app, creates account (name, email, create password)	Collects baseline data, sets up personalised profile, explains how the app works in 3 screens
2. First Analysis	Opens camera, takes a picture of their meal OR Uploads meal picture from gallery and taps 'Start Analysis'	Sends picture to AI model, classifies food groups, generates the first nutrition report
3. Feedback	Reads nutrition report, sees food groups breakdown	Displays results clearly, explains what each food group does, highlights missing groups
4. History	Opens the History tab, searches or filters meals by week or month, and reviews logged entries	Displays meal history with timestamps and categories, updates the view based on selected filters, and keeps records organized for easy tracking
5. Settings	Opens Settings to personalize preferences and manage notifications	Applies changes instantly and confirms updated preferences

5.2 Feature Breakdown

FEATURE 1 - Onboarding Flow

Feature	Onboarding Flow
Description	A 4-screen onboarding experience that collects user profile data, explains the app's purpose, and sets expectations before the first analysis.
Screens & UI	Screen 1: Welcome screen with PlateLensAI logo, tagline ('Upload meals - Get instant analysis and tips'), and 'Get Started' CTA button.Screen 2: Account creation - full name, email, create password. .Screen 3: How it works - 3-step illustration: (1) Take or

	Upload a picture, (2) Get your food group breakdown, (3) Build better habits. 'Start Analysis' CTA.
Input Fields / Data Collected	Full name (text input, required), Email address (text input, required, validated), Create password (text input, required, min 8 characters).
Conditions for success	User completes all required fields and lands on the home screen. Account is created in Firebase Auth. Profile data stored in user database. Onboarding screens are never shown again after completion, the login page shows for a returning user.

FEATURE 2 - Meal Picture Analysis (Core AI Feature)

Feature	Meal Picture Analysis
Description	The core feature. User takes or uploads a picture of their meal and the app sends the image to the AI classification model, which identifies the food groups present and returns a structured result.
Screens & UI	Home Screen: Large circular 'Start Analysis' CTA button in the centre. Camera icon. 'Take picture' button. 'Choose File' button Option to upload from gallery...,Processing Screen: Animated PlateLensAI logo with loading indicator and message: 'Analysing...' - max 5 seconds. Result Preview: Thumbnail of captured picture with detected food items overlaid as labels and analysis summary displayed. 'Analyze again' option if the result seems wrong.
Input Fields / Data Collected	Meal picture (camera capture or gallery upload - JPG/PNG, max 5MB), Timestamp (auto-captured), Meal type selection (Breakfast / Lunch / Dinner / Snack - shown after analysis, preferred).
Conditions for success	AI model returns classification result within 5 seconds in 95% of cases. At least one food group is identified in the analysis. Classification confidence score $\geq 70\%$ (below this threshold, app shows a 'We are not fully sure - please retake' message). Picture is stored in user's meal history, if toggled in settings. Result is linked to correct meal type and timestamp.For the analysis summary, the app must correctly identify food groups present and missing in each analyzed meal with high accuracy.

FEATURE 3 - Nutrition Feedback Report

Feature	Nutrition Feedback Report
Description	The personalised nutrition report generated after each analysis and summary given. Displays tips on what food group(s) is or are missing to balance the meal.
Screens & UI	Tips Screen - Back button at the top left. Title displayed as Tips to Balance Your Meal. The screen shows one continuous paragraph of guidance based on the analyzed meal. It explains what food groups are missing and gives examples of what to add.
Input Fields / Data Collected	No additional input required. All data is derived from the AI classification result.
Conditions for success	The feedback must be actionable and culturally relevant with clear examples and portion guidance that users can realistically follow. The Tips screen must display information in a simple continuous format that users can understand without confusion. The system must encourage users by showing their current meal status and motivating them to improve with specific suggestions. The app must integrate seamlessly with history logging and reporting so that users can track progress over time. The experience must be smooth and responsive with minimal errors or delays during analysis, logging, and feedback generation.

FEATURE 4 - Meal History Log

Feature	Meal History Log
Description	A chronological log of all meals analyzed, organised by date, with the ability to review past nutrition reports.
Screens & UI	History Screen. Scrollable list of past meals. Each row shows a meal picture thumbnail, meal type, date and time. Filter Bar. Filter options are limited to date range only with choices for this week, this month, and all. Meal Detail Screen. Tapping any meal in the list opens the full nutrition feedback report for that meal.
Input Fields / Data Collected	No new input - all data pulled from existing meal log. Date range filter (date picker).
Conditions for success	History loads within 2 seconds. All previously analyzed meals are displayed in reverse chronological order. Filters work correctly and return accurate results. Tapping a meal always opens the correct corresponding report. Empty state (no meals yet) shows an illustration with a text.

FEATURE 5 - Preferences and Push Notification Reminders

Feature	Preferences and Push Notifications
Description	Allows users to manage preferences such as saving analyzed meals to history and auto tagging meal time. Provides notification controls where users can toggle meal analysis reminders and sound or vibration alerts.

Screens & UI	The Interface shows toggles for preferences and notifications. Includes storage and data options such as clear analysis history and clear cache or temporary files. In one version the screen also shows a user profile section with avatar, name, email and a change password link. In another version the screen includes system information options such as about application, help or FAQ and a red delete my account option.
Input Fields / Data Collected	Notification permission is requested during onboarding and users can later adjust notification preferences in the settings screen by toggling each type on or off.
Conditions for success	Notifications are only sent if the user has granted permission. Daily reminders are not sent on days when a meal has already been analyzed. Streak or missing food group nudges are sent only when conditions are met. Notification tap always opens the relevant screen such as camera, or history. Notification opt out is respected within one hour of preference change.

6 . USER STORIES

6.1 Motivation

- As a Nigerian university student, I want to take a picture of my meal, get it analyzed and instantly know which food groups it contains, so that I can understand if my diet is balanced without needing nutritional expertise.
- As a student who eats campus canteen food daily, I want the app to recognise Nigerian meals like egusi soup and jollof rice, so that I feel the feedback is relevant to my actual life.

6.2 Retention

- As a returning user, I want to see my meals analysis history over the past week, so that I can track whether my eating habits are improving.
- As a user who sometimes forgets to analyze my meals, I want a gentle reminder notification if I haven't logged a meal by afternoon, so that I stay consistent without feeling pressured.

6.3 Reward

- As a user who has analyzed meals consistently for a week, I want to receive a streak celebration, so that I feel motivated to continue building the habit.

6.4 Recovery

- As a user who missed several days of analysis, I want the app to welcome me back warmly rather than making me feel guilty, so that I feel comfortable re-engaging without abandonment shame.

7 . USER ACCEPTANCE CRITERIA (UAC)

Feature	Acceptance Criteria	Pass condition
Meal Picture	User captures or uploads a picture of a Nigerian meal	AI returns classification result within 5 seconds with ≥ 1 food group identified
AI Classification Accuracy	AI model classifies common Nigerian meals correctly	$\geq 85\%$ accuracy on internal test dataset of 500 Nigerian meal images
Nutrition Feedback Report	Report is generated after every successful analysis	All 6 food group cards display, score calculates correctly, suggestions are relevant

Meal History	User can view all past analysis in chronological order	All meals display correctly, filters work accurately, tapping opens correct report
Push Notifications	User receives reminders only if opted in and criteria met	No notification sent on days with existing analysis, opt-out respected within 1 hour
Onboarding	New user completes profile setup before first analysis	Account created, profile stored, onboarding never shown again after completion

8 . FUNCTIONAL REQUIREMENTS

8.1 Must Haves (P0)

- User account creation, authentication, and profile management
- Meal picture capture via device camera
- Gallery upload as alternative to camera capture
- AI food group classification (MobileNetV2 model, 6 food groups)
- Nutrition feedback report with food group breakdown
- Personalised meal suggestions based on missing food groups

8.2 Should Haves (P1)

- Meal history log with chronological display
- Preferences and Push notification system: daily reminder and weekly summary

8.3 Nice to Haves (P2)

- Weekly food group coverage dashboard
- Social sharing - generate shareable nutrition score card image

9 . NON - FUNCTIONAL REQUIREMENTS

Requirement	Specification	Rationale
Performance	Analysis-to-result time ≤ 5 seconds in 95% of cases. App launch time ≤ 3 seconds	Students on mobile data expect instant results. Delay beyond 5s causes abandonment
Scalability	Backend must support 10,000 concurrent users without degradation	University students share content virally. Launch could drive rapid user spikes
Security	All user data encrypted in transit (HTTPS/TLS 1.3) and at rest. Passwords hashed with encrypt	User dietary data is personal health information and must be treated as sensitive
Accessibility	Minimum font size 14sp. Colour contrast ratio $\geq 4.5:1$. VoiceOver/TalkBack compatible for all core screens	Students with visual impairments or reading difficulties must not be excluded
Offline Capability	App must display cached history and dashboard data when offline. Analysis feature requires internet	Campus Wi-Fi is unreliable at Nigerian universities. Offline viewing prevents frustration
Data Privacy	No meal picture is stored on device beyond 24 hours. Cloud storage opt-in only. NDPR compliant	Nigeria Data Protection Regulation (NDPR) applies. Students must consent to cloud image storage
Model Reliability	AI model must return a valid result or a graceful error in 100% of requests	Silent failures destroy trust. Every failed classification must show a clear, friendly error message
Battery Usage	App must not consume more than 5% battery per 10-minute active session	Students on low-budget Android devices are battery-sensitive. Excessive drain causes uninstalls

10. EDGE CASES

Edge Case	Scenario	Handling
Low confidence analysis	AI returns confidence score below 70%	Show message: 'We are not fully confident about this result - try retaking in better lighting.' Option to proceed anyway or analyze again.
Unrecognisable food	Picture contains non-food objects or a food not in training data	Show message: 'We couldn't identify this meal. Please try a clearer picture' Log as 'Unknown meal' in history.
Multiple dishes in one picture	Student pictures a full tray with 4+ dishes	Model classifies the full image holistically. Report states: 'This result covers all visible dishes in your picture.' No per-dish breakdown in v1.
Poor lighting / blurry picture	picture is too dark or out of focus for classification	Image quality check before sending to model. If below threshold, show: 'Picture quality is too low. Please retake in better lighting.'
No internet connection during analysis	User attempts to analyze while offline	Show message: 'You need an internet connection to analyse your meal. Your picture has been saved, tap to analyse when connected.'
User analyzes the same meal twice	Duplicate analysis within 15 minutes of same meal type	Show prompt: 'It looks like you already analyzed a meal recently. Add this as a new meal or replace the previous analysis?'
App crash during analysis	App crashes while AI is processing	On relaunch, check for incomplete analysis session. If found, show: 'It looks like your last analysis didn't complete. Would you like to retake it?'
First-time user with no data	Dashboard opened before any meals analyzed	Show illustrated empty state: 'Your nutrition journey starts with your first analysis. Analyze a meal!' with CTA button

11 . EVENT TRACKING SHEET

Event name	Trigger	Properties	Purpose
onboarding_started	User opens app for first time	device_type, os_version	Track funnel start
onboarding_completed	User reaches home screen	time_to_complete, university	Track completion rate
analysis_initiated	User taps 'Analyze Meal' button	source (camera/gallery), meal_type	Track analysis intent
analysis_completed	AI returns result	confidence_score, food_groups_detected, scan_duration_ms	Core engagement metric
analysis_failed	AI returns error or low confidence	error_type, confidence_score	Track model failures
report_viewed	User opens nutrition feedback report	nutrition_score, food_groups_present	Track report engagement
manual_override_used	User corrects AI result	original_group, corrected_group	Model improvement signal
history_viewed	User opens meal history tab	filter_applied, meals_in_view	Track retention behaviour
dashboard_viewed	User opens weekly dashboard	week_selected, weekly_score	Track habit loop
notification_opened	User taps push notification	notification_type, time_since_last_scan	Track re-engagement
streak_milestone_reached	User achieves 3, 7, 14, 30-day streak	streak_length	Track retention depth

12 . ASSUMPTIONS & RISKS

12.1 Assumptions

- Users have smartphones with functional cameras and at least 2G internet connectivity
- The MobileNetV2 model trained on the existing Nigerian food dataset achieves $\geq 85\%$ accuracy at launch
- Nigerian university students are willing to take pictures of their meals before eating if the benefit is clear
- React Native performance on mid-range Android devices (the dominant Nigerian student device) is sufficient for the camera and AI features
- Firebase will be used for authentication and real-time database in v1

12.2 Risks

Risk	Likelihood	Mitigation
AI model accuracy below 85% at launch	Medium	Expand training dataset before launch. Implement manual override to capture correction data for continuous model improvement
Poor camera quality on low-end Android devices	High	Implement image quality pre-check before sending to model. Provide clear in-app guidance on how to take a good picture
Low user retention after first analysis	Medium	Invest in push notification strategy and streak mechanics. Run 30-day retention analysis in beta before full launch
NDPR compliance issues with storing meal pictures	Low-Medium	Make cloud picture storage opt-in. Implement automatic local deletion after 24 hours. Obtain legal review of privacy policy before launch
Model performing poorly on mixed-dish pictures	Medium	Train on multi-dish images in v1.1. Communicate clearly in v1 that single-dish pictures give best results
Network failures during analysis	High (campus data)	Implement offline scan queue - save picture locally and process when connectivity is restored

13 . CONSTRAINTS

- Budget: Early-stage startup budget. Must prioritise free-tier cloud services (Firebase, Cloudinary free tier for image storage, Python FastAPI on a basic server) until paid user base is established.
- Team size: Small team of 2 - 3 people across product, design, and engineering for v1. Scope must be tightly controlled, no features beyond P0 and P1 until v1 is stable.
- Timeline: MVP must be ready for beta testing within 12 weeks of development start. Full launch within 16 weeks.
- Regulatory: Must comply with Nigeria Data Protection Regulation (NDPR). No personal health data sold or shared with third parties. Privacy policy must be reviewed by a Nigerian legal practitioner before launch.
- Device support: Must support Android 8.0+ and iOS 13+. Must be optimised for devices with 2GB RAM; the most common student device category in Nigeria.
- Language: English only for v1. Yoruba, Igbo, and Hausa localisation considered for v2 based on user demand.

14 . OUT OF SCOPE - 1.0

- Calorie counting or macro tracking is not in scope. PlateLensAI focuses on food group balance, not calorie deficit/surplus.
- Personalised meal plans or recipe suggestions will be considered for v2 after sufficient user data is collected.
- Integration with wearables (Fitbit, Oura, etc.) for future release.
- Social features such as community feed, friend comparisons, leaderboards will be for future release.
- Web version for mobile-only for v1.
- Institutional dashboard for universities or cafeteria operators for future B2B feature.
- Paid premium tier for monetisation strategy to be defined post product-market fit validation.

15 . TENTATIVE DEVELOPMENT TIMELINE

Task	Design Ready	Development Starts	Beta Launch	Full Launch
Leadership Approval	Week 0	—	—	—
Onboarding Flow	Week 1	Week 2	Week 6	Week 10
Meal Picture Analysis + AI Integration	Week 2	Week 3	Week 7	Week 11
Nutrition Feedback Report	Week 2	Week 3	Week 7	Week 11
Meal History Log	Week 3	Week 4	Week 8	Week 12
Preferences and Push Notifications	Week 5	Week 6	Week 10	Week 14
Beta Testing & Bug Fixes	—	—	Week 7–10	—
Full Public Launch	—	—	—	Week 16

16 . DEPENDENCIES

16.1 Infrastructure Requirements

- Python FastAPI server for AI model hosting - minimum 2GB RAM, GPU-optional (CPU inference acceptable for v1)
- Firebase Authentication and Firestore Database
- Cloudinary or Firebase Storage for meal image storage
- Push notification service - Firebase Cloud Messaging (FCM)

16.2 Budget Approvals

- Cloud server costs - estimated ~~₦15,000–₦30,000~~/month for v1 scale
- App Store developer accounts - Apple (\$99/year), Google Play (\$25 one-time)
- Legal review of privacy policy - ~~₦50,000–₦100,000~~ one-time

16.3 Partner Support

- No third-party nutrition APIs required - all data is generated internally by the AI model
- Nigerian food nutritional database - requires partnership with NAFDAC or National Nutrition Institute for validation of food group data accuracy

16.4 Internal Dependencies

- Engineering: Backend API (Python FastAPI), React Native mobile app, AI model deployment pipeline.
- Design: All screen wireframes and design systems must be completed before engineering starts each feature.
- Marketing: University ambassador programme and launch campaign must begin 4 weeks before beta launch.
- Compliance: Privacy policy and NDPR compliance review must be completed before beta launch.

17 . RELATED DOCUMENTS

Document	Owner - Location
Engineering Planning Document	Engineering Lead - [Link to be added]
Design Planning Document & Figma File	Design Lead (Inioluwa Adeye) - [Figma link to be added]
Go-to-Market Planning Document	Marketing Lead - [Link to be added]
AI Model Architecture & Training Report	Engineering Lead - [Link to be added]
Privacy Policy & NDPR Compliance Review	Legal - [Link to be added]
User Research Interview Notes	Product Lead (Inioluwa Adeye) - [Notion link to be added]
Beta Testing Feedback Log	Product Lead - [To be created by Week 7]